The Limits to Growth: Sustainability and the Circular Economy

Lecture 5: Circular Economy

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- Updated versions of these slides will be available in our Github repository.

INTRODUCTION

Introduction

Sustainability – Definition

„Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”

Report of the World Commission on Environment and Development: Our Common Future (1987) – http://www.un-documents.net/our-common-future.pdf

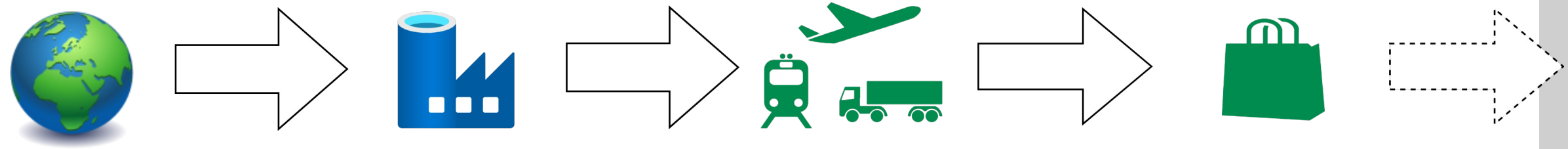
Introduction

Sustainability – Implications

Sustainability → Consume less

Introduction

Linear (Industrial) Economy



RESOURCES **MANUFACTURING** **DISTRIBUTION** **POINT OF SALE**

GLOBALIZED INDUSTRIALIZED PROCESSES

NATURAL RESOURCES (Water, Energy, Materials) and Human Resources (Labor and Skills) are used to produce manufactured **objects** (infrastructure, buildings, equipment, durable projects).

GLOBALIZED DISTRIBUTION LOGISTICS

COMMERCIALIZATION at the factory gate or at point of sale these objects are put to use (sold, leased, rented, sold as service)...

Image adapted from: Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Introduction

Linear Economy

Introduction
Linear Economy

"Take – Make – Dispose"

Introduction

Linear (Industrial) Economy – Definition

"Its objectives are to **maintain value** (not to create value added), to **optimise stock** management (not flows) and to **increase the efficiency of using goods** (not of producing goods)"

Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Introduction

Linear (Industrial) Economy – Waste Management as last Step

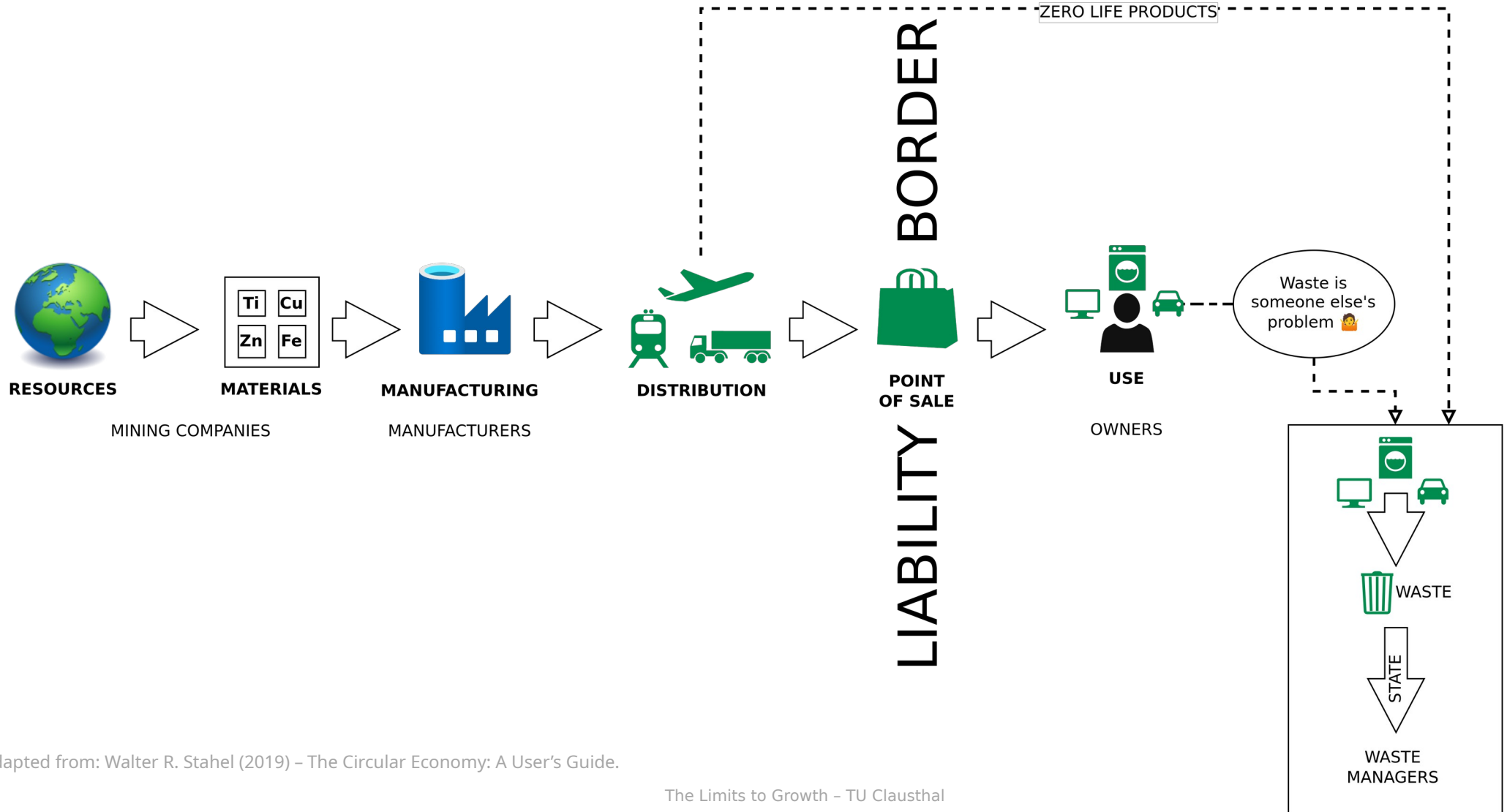


Image adapted from: Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Introduction

Horrible Waste Management





1. "Landfill at Upernavik" by ulalume – https://www.flickr.com/photos/96649248@N00/43867280734 – CC BY-NC-ND 2.0.
2. Christian Hüpfer – https://flic.kr/p/aKXw2F – CC BY-SA 2.0.

Introduction

Horrible Waste Management – Nuclear Waste





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Introduction

Horrible Waste Management – Nuclear Waste

→ 2 generations profited from cheap nuclear energy
→ 40,000 generation will have to live with the waste

https://www.spiegel.de/wissenschaft/technik/endlager-suche-was-wie-viel-und-wohin-damit-die-deutsche-atommuell-bilanz-a-7a153ba3-029a-4e55-adaf-3312b7427d9e

Introduction

Horrible Waste Management – Plastic



"Pollution Plastique" by Mouenthias is licensed with CC BY-SA 4.0. To view a copy of this license, visit https://creativecommons.org/licenses/by-sa/4.0/

Introduction

The Limits to Recycling

- Recycling often requires a lot of energy
- Some materials cannot be recycled at all (yet)
- Impurities are challenging
- Often requires a lot of manual labor
- Recycled material often with lower quality than input material
- False sense of safety!

I am <u>not</u> saying you should stop recycling!

→ Recycling is great but it is better to make sure that we do not have to recycle anything.

→ Buying less (e.g., only the essentials) is way more effective.

“We buy things we don't need, to impress people we don't like.” - Tyler Durden / Chuck Palahniuk

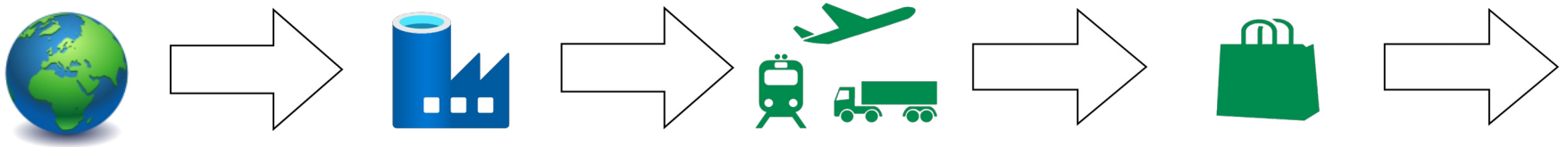
Waste = Inefficiency

Introduction

From Linear to Circular

LINEAR INDUSTRIAL ECONOMY

PRODUCERS *IN-CONTROL*



RESOURCES | **MANUFACTURING** | **DISTRIBUTION** | **POINT OF SALE**

GLOBALIZED INDUSTRIALIZED PROCESSES

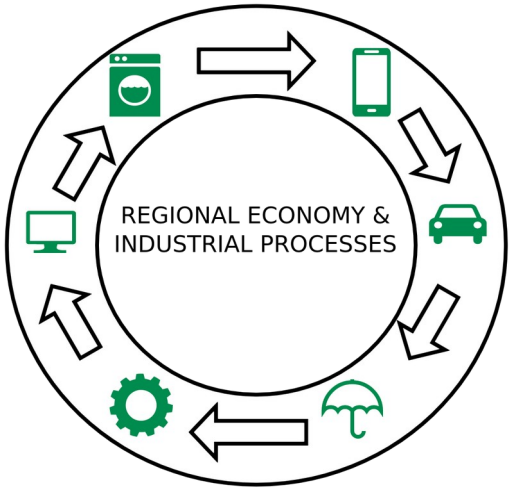
NATURAL RESOURCES (Water, Energy, Materials) and Human Resources (Labor and Skills) are used to produce manufactured **objects** (infrastructure, buildings, equipment, durable projects).

GLOBALIZED DISTRIBUTION LOGISTICS

COMMERCIALIZATION at the factory gate or at point of sale these objects are put to use (sold, leased, rented, sold as service)...

CIRCULAR ECONOMY

USERS *IN-CONTROL*



... and later **REUSED** as objects **AT THEIR HIGHEST VALUE** and function, or recovered as molecules at their highest value and purity

Image adapted from: Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Introduction

Linear Economy – Main Challenges

- Environmental pollution / waste
- Waste of resources
  - → Just recycling is not gonna do it

Introduction

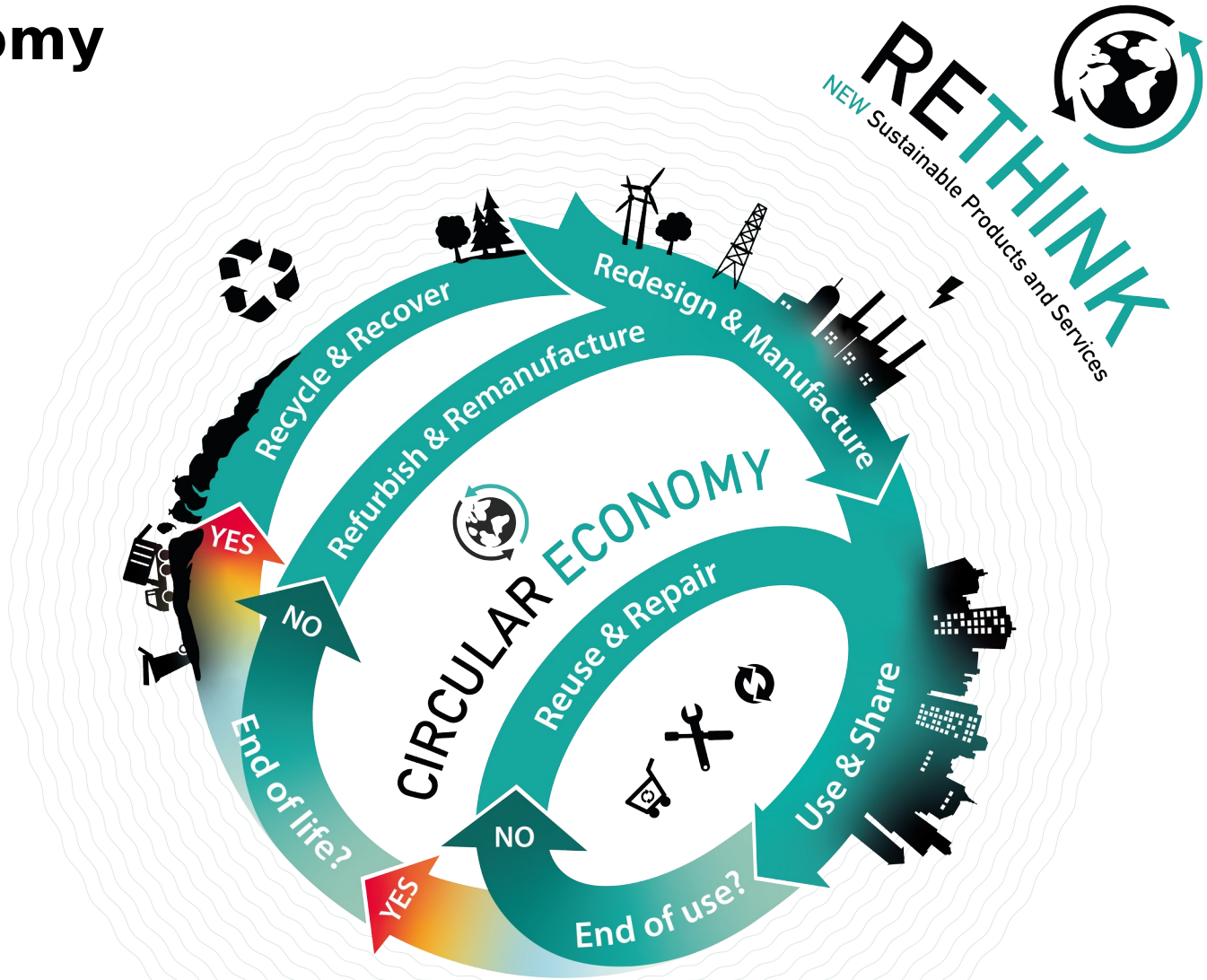
Linear Economy – Main Challenges

- Environmental pollution / waste
- Waste of resources
  - → Just recycling is not gonna do it

- Therefore:
  - Consume less
  - Make stuff last longer
  - Maximize resource utilization

THE CIRCULAR ECONOMY

Circular Economy



Center for Digital Technologies (DIGIT) (2022) – https://www.digit-research.de/fileadmin/DIGIT/Presse_und_News/Downloads/Grafikpaket_Reallabor_DCE_Kreislaufmodell_V2_ccby.zip – CC BY 4.0.

Circular Economy

Definition

“Conceptualizing the circular economy: An analysis of 114 definitions.”

Kirchherr, Julian, Denise Reike and Marko P. Hekkert. Resources Conservation and Recycling 127 (2017): 221-232.

https://doi.org/10.1016/J.RESCONREC.2017.09.005

Circular Economy

Definition

"A circular economy is an industrial system that is restorative or regenerative by intention and design. It replaces the 'end-of-life' concept with restoration, shifts towards the use of renewable energy, eliminates the use of toxic chemicals, which impair reuse, and aims for the elimination of waste through the superior design of materials, products, systems, and, within this, business models."

– Ellen MacArthur Foundation

https://www.ellenmacarthurfoundation.org/assets/downloads/publications/Ellen-MacArthur-Foundation-Towards-the-Circular-Economy-vol.1.pdf

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– Ellen MacArthur Foundation

"The circular economy is a model of production and consumption, which involves sharing, leasing, reusing, repairing, refurbishing and recycling existing materials and products as long as possible. In this way, the life cycle of products is extended."

– European Parliament

https://www.ellenmacarthurfoundation.org/assets/downloads/publications/Ellen-MacArthur-Foundation-Towards-the-Circular-Economy-vol.1.pdf
https://www.europarl.europa.eu/news/en/headlines/economy/20151201STO05603/circular-economy-definition-importance-and-benefts

Circular (Industrial) Economy

Definition

"The circular industrial economy **manages stocks** of manufactured assets, such as infrastructure, buildings, vehicles, equipment and consumer goods, to **maintain their value and utility as high as possible for as long as possible**; and stocks of **resources at their highest purity and value**."

Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Circular Economy

Sustainability – Definition

„Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”

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Circular Economy

Characteristics

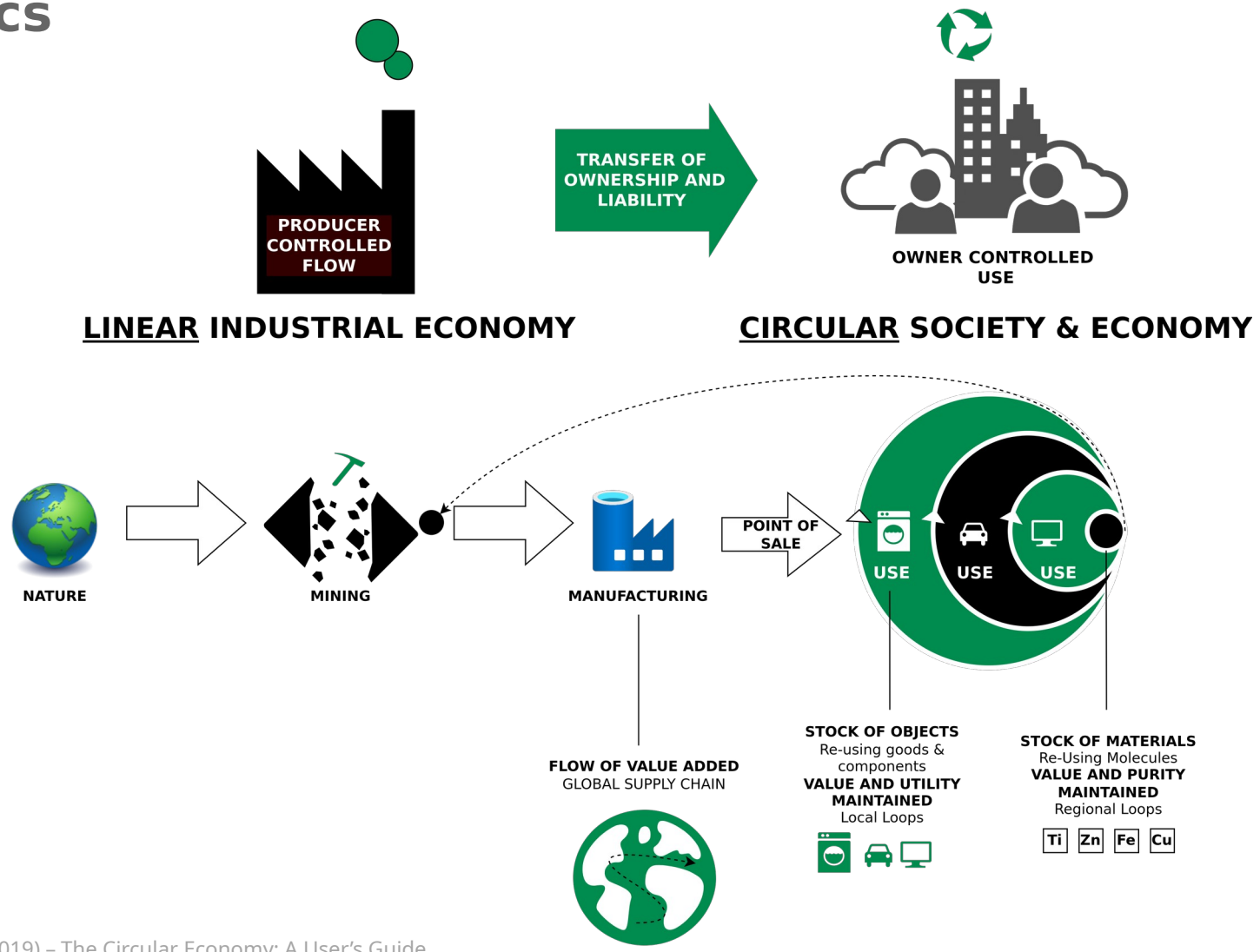


Image adapted from: Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Circular Economy

The Era of R

Techno-commercial strategies to keep goods and components at highest value level through:

- **R**euse
- **R**epair
- **R**emarket
- **R**emanufacture
- **R**e-refine
- **R**eprogramme goods

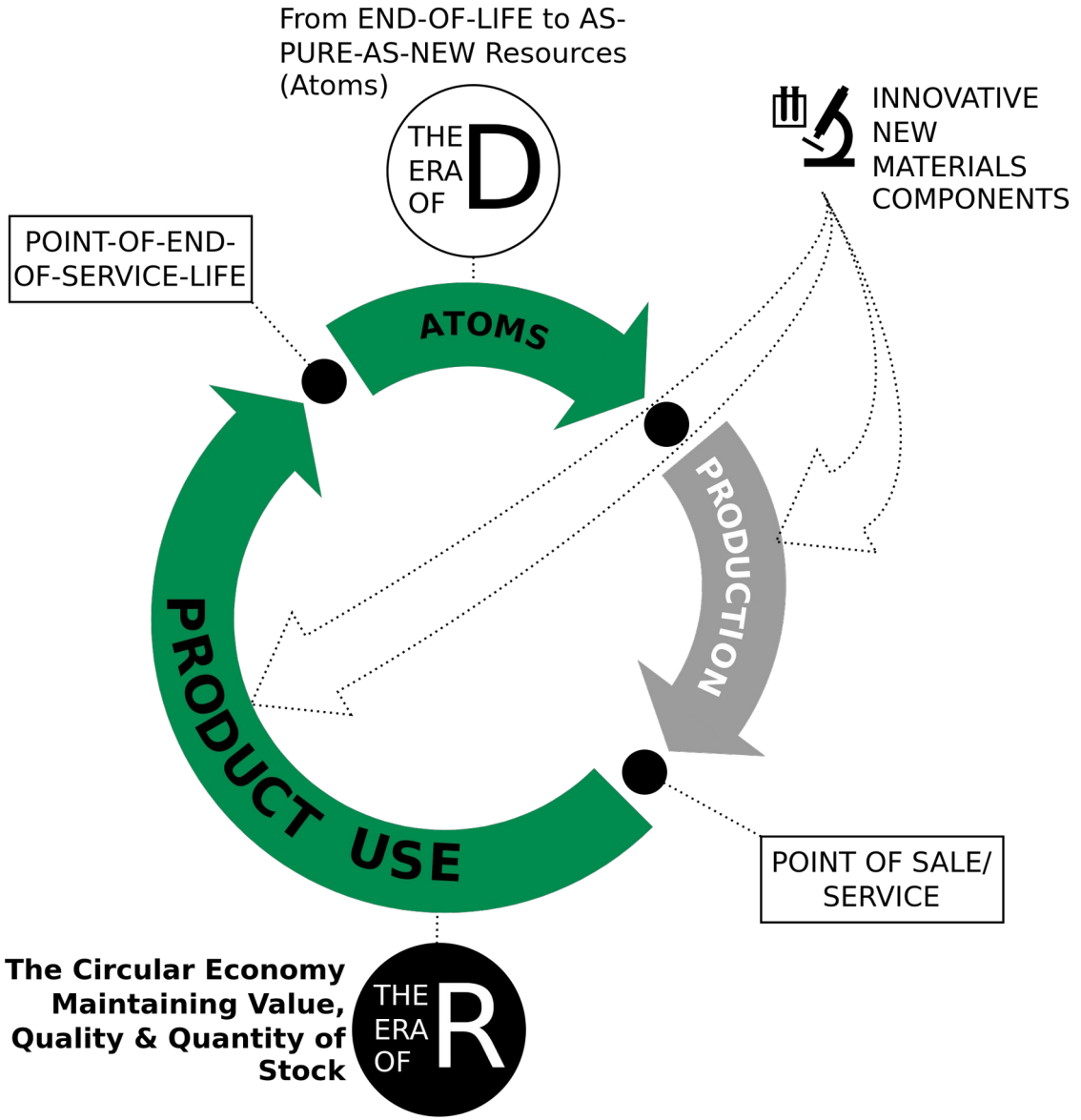


Image adapted from Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Circular Economy

The Era of D

Technologies and actions to recover atoms and molecules at highest quality (purity and value) level as pure as virgin:

- **D**e-polymerise
- **D**e-alloy
- **D**e-laminate
- **D**e-vulcanise
- **D**e-coat materials
- **D**e-construct high-rise buildings and major infrastructure

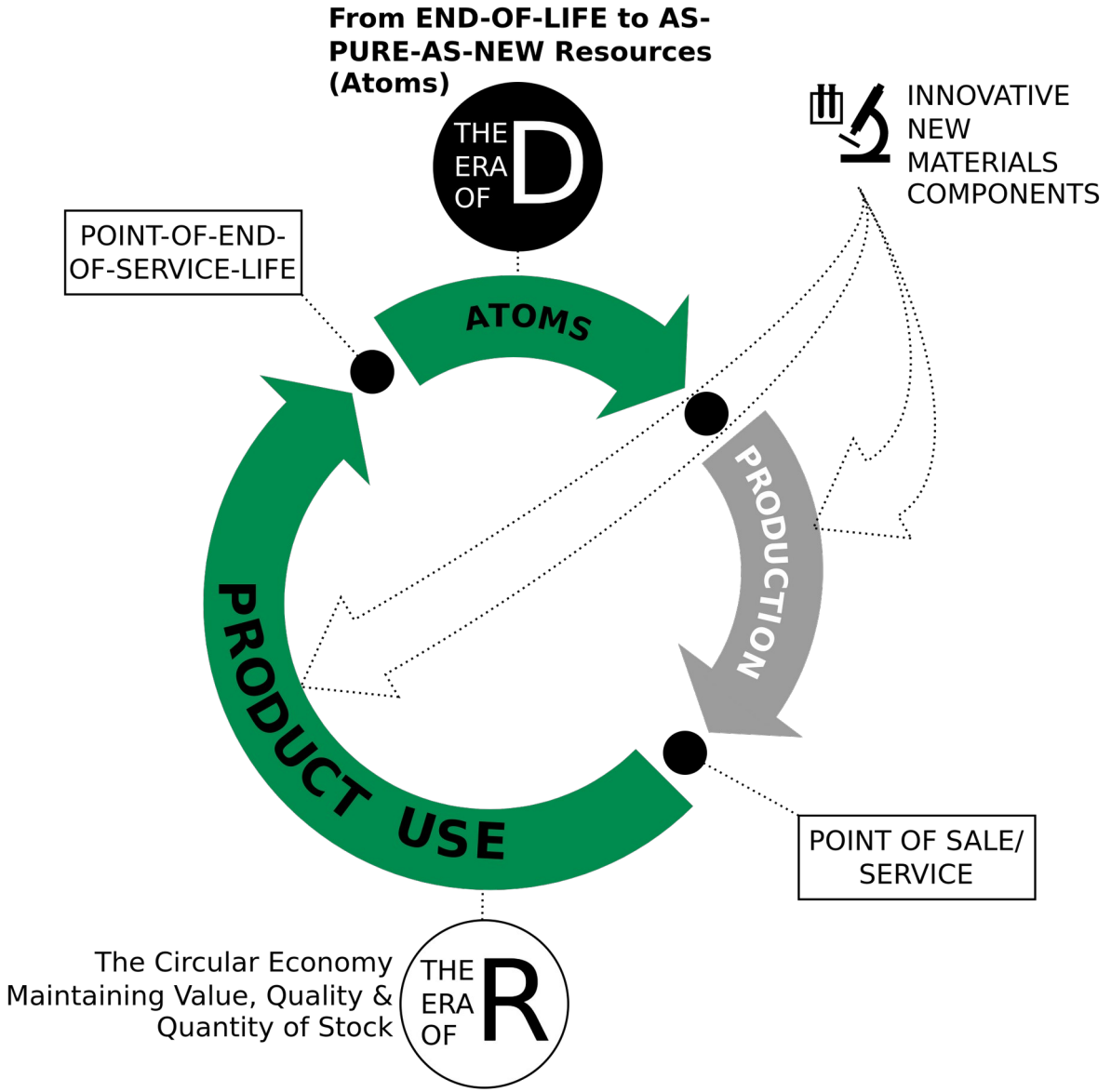


Image adapted from Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Circular Economy

End-of-Life – Reuse or Recycle?

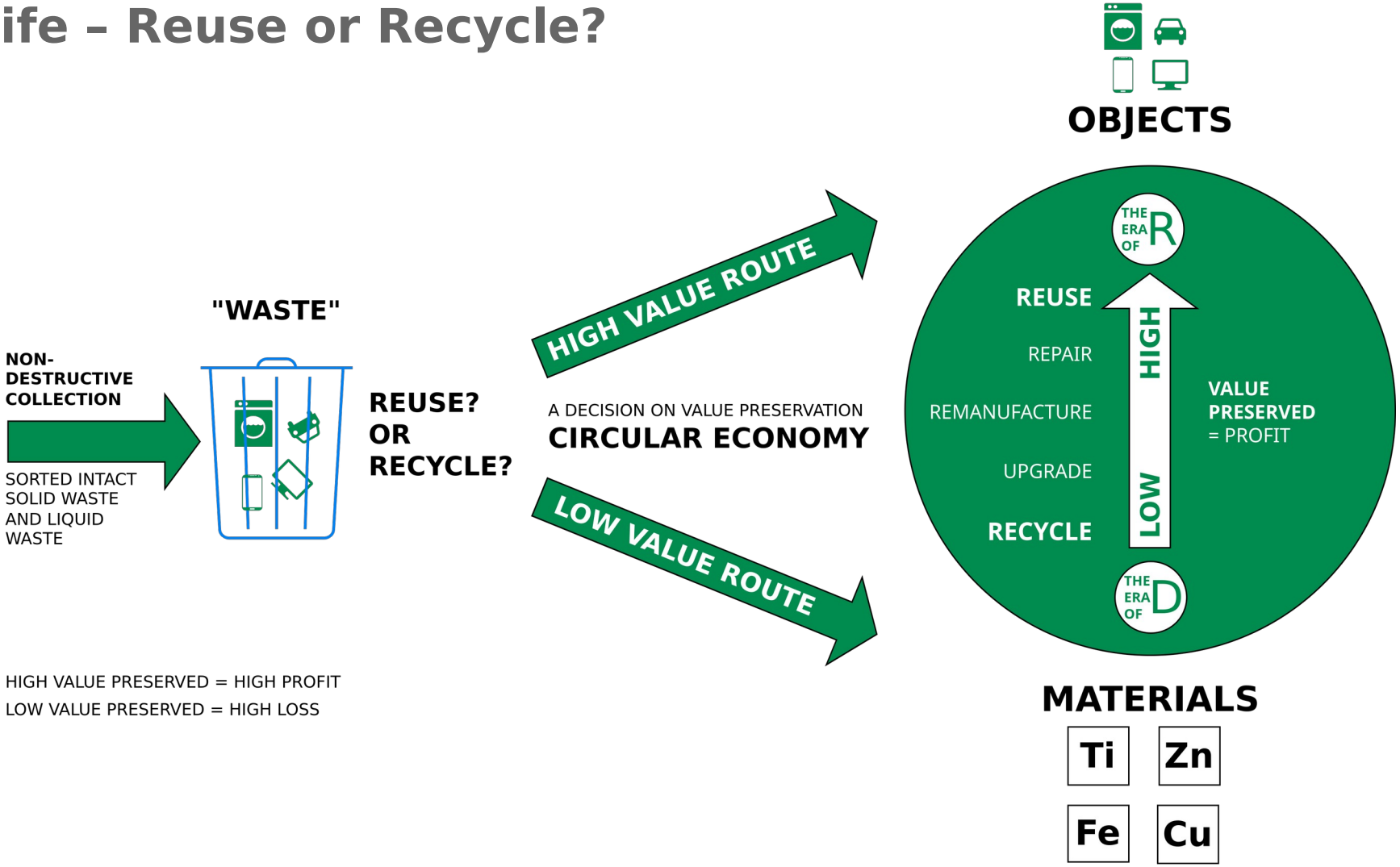


Image adapted from Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Circular Economy

More Wealth/Jobs From Less Resource Consumption

- **Linear Economy:** Low hr/kg (labor input per weight) ratios, coherent with mass production in highly mechanized processes, and low to medium €/kg (value per weight) ratios, in a range from basic materials like cement to smart goods like USB memory sticks

- **Circular Economy:** Higher hr/kg and €/kg ratios for reuse, remanufacture and selling performance (goods as a service), in a group with new technologies, such as life sciences and nanotechnologies, which by nature produce dematerialized objects.

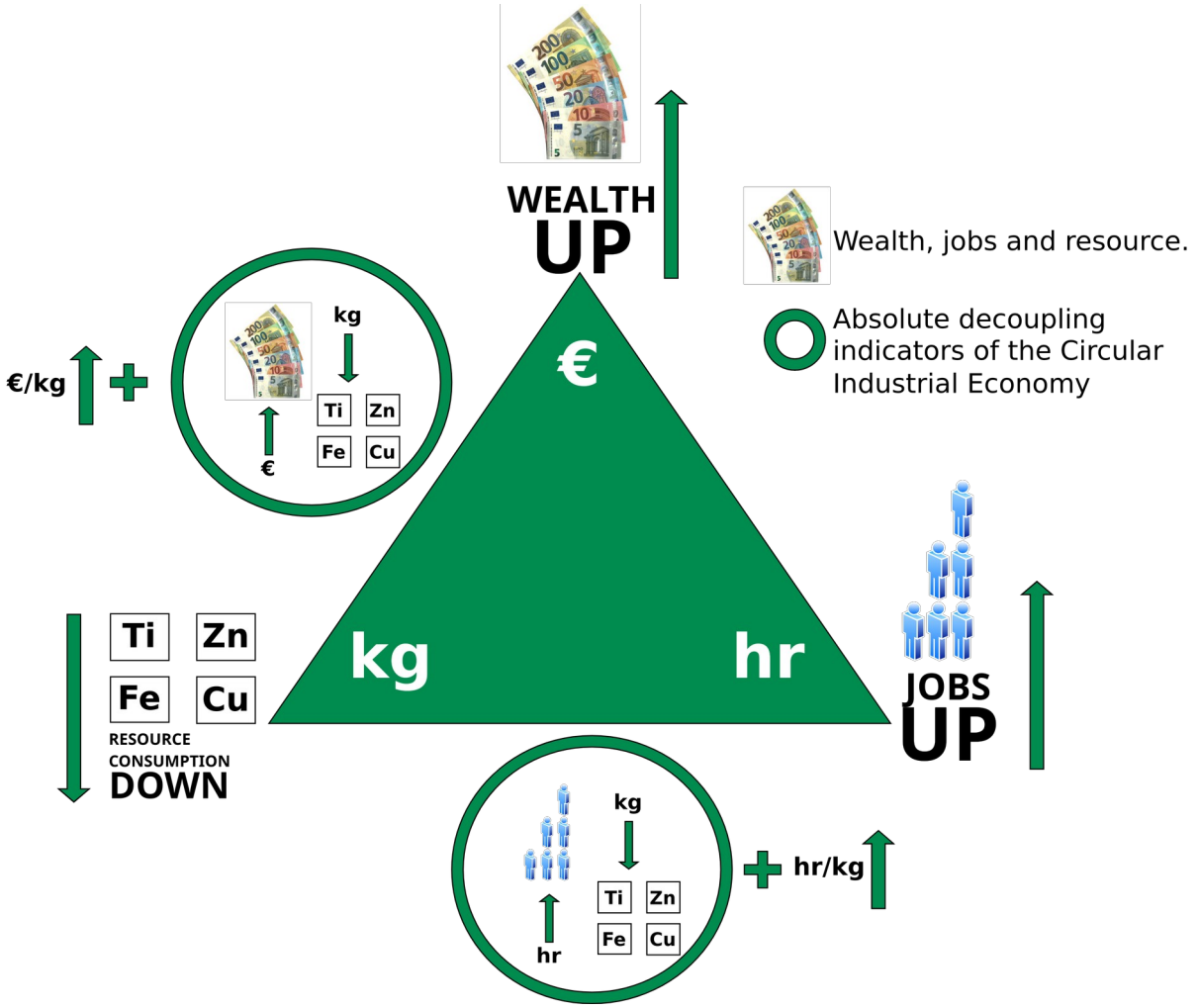


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Circular Economy

New Engine vs. Remanufacturing A Car Engine

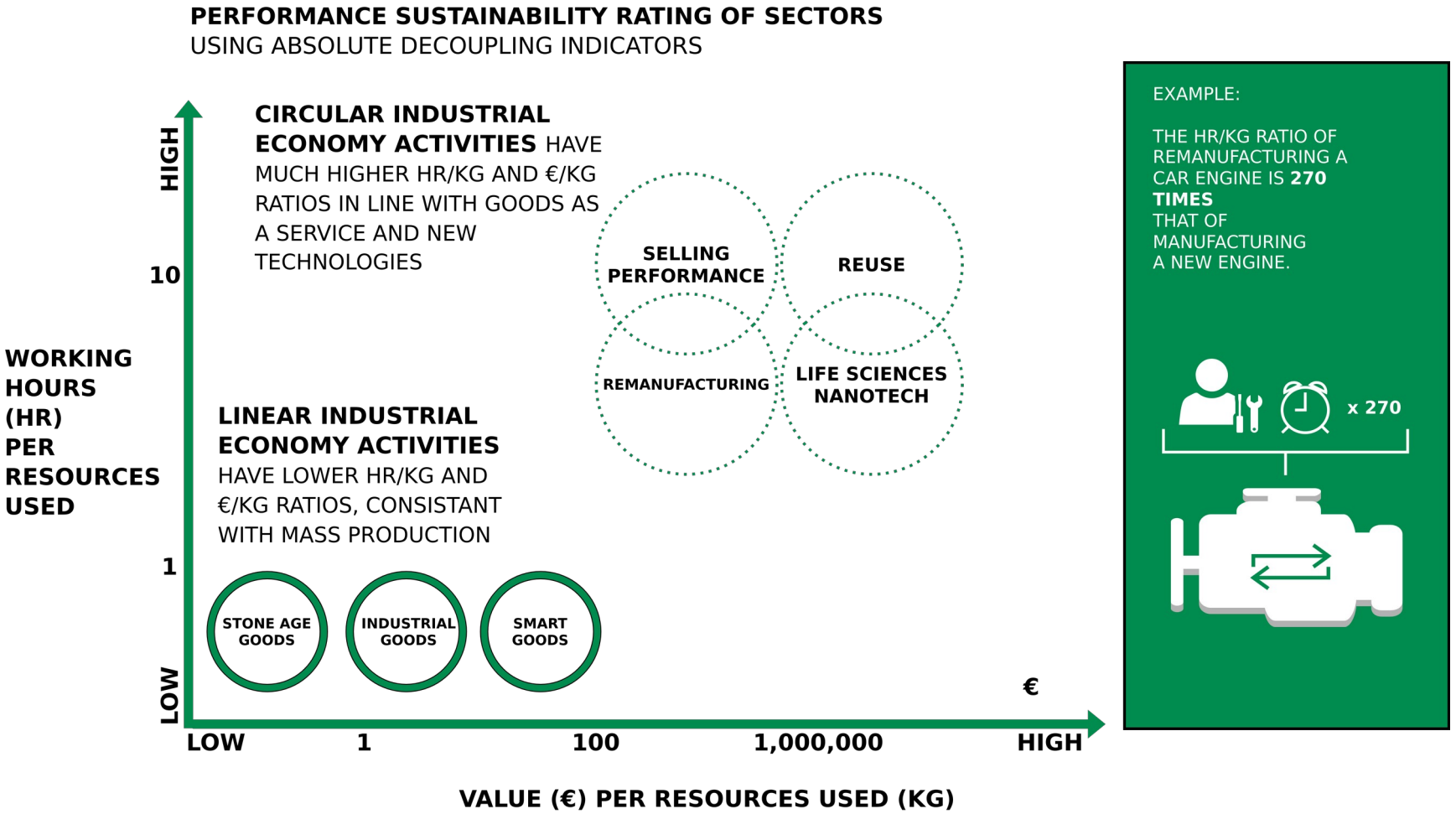


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Circular Economy

Business Model Characteristics

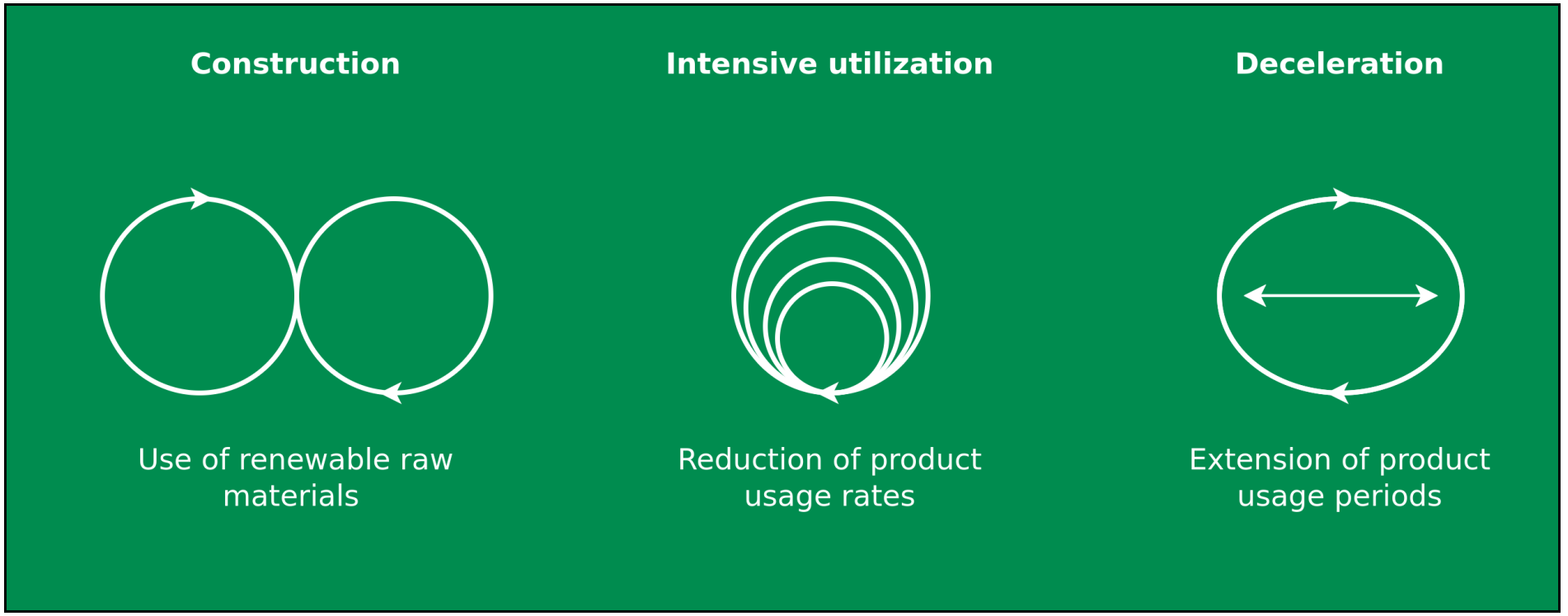


Image adapted from: M. Jaeger-Erben, F. Hofmann (2019) – Kreislaufwirtschaft - Ein Ausweg aus der sozial-ökologischen Krise? – Link.

Circular Economy

BUSINESS MODEL TOPOLOGY OF THE CIRCULAR ECONOMY

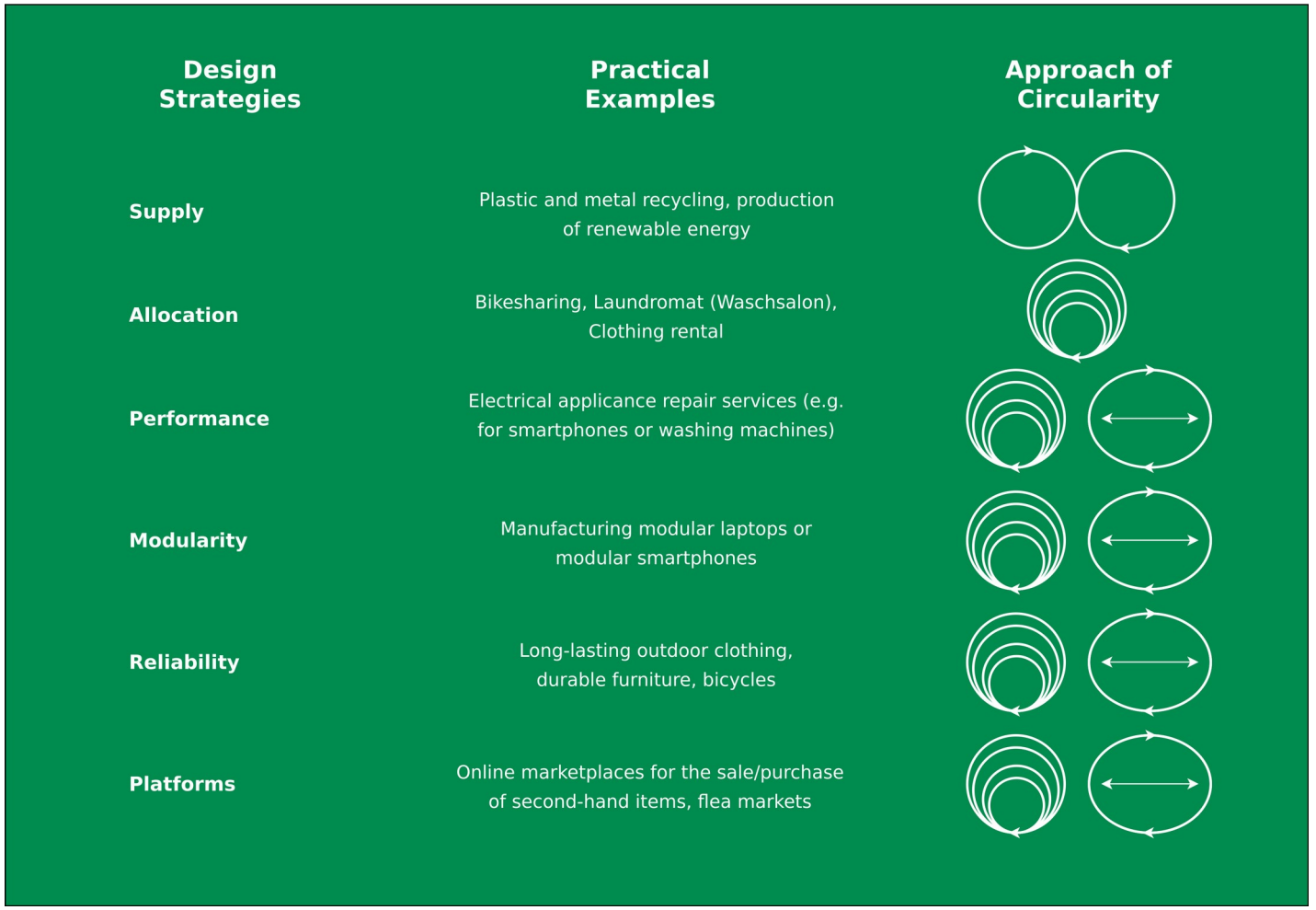


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PERFORMANCE ECONOMY

Performance Economy

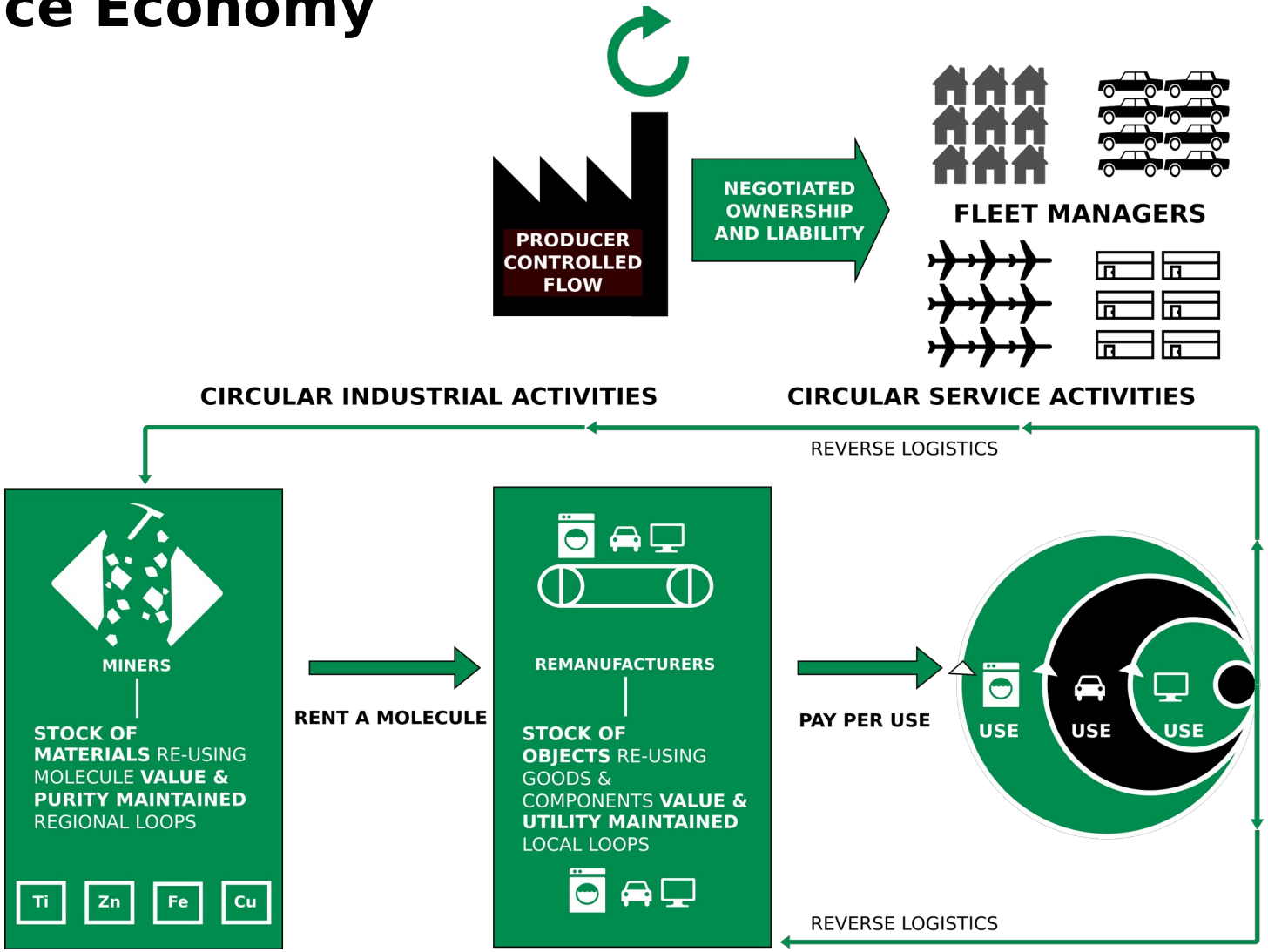


Image adapted from Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Performance Economy – Definition

„**The Performance Economy sells results instead of objects.** Its economic actors may be manufacturers of durable objects or fleet managers operating them. In both cases, they sell the use of these objects as a service over the longest possible period of time and maximize their profits by exploiting both efficiency and sufficiency solutions. “

Walter R. Stahel (2019) – The Circular Economy: A User’s Guide.

Performance Economy – Most sustainable CE business model?

- Stahel argues:
  - “The Performance Economy of selling goods and molecules as a service, function guarantees or results and performance, is the most sustainable business model of the circular industrial economy because by internalising the costs of product liability, of risk and waste, it offers manufacturers a strong financial incentive to prevent losses and waste.”

Walter R. Stahel (2019) – The Circular Economy: A User’s Guide.

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  - "It maximises the profit potential by exploiting sufficiency, efficiency and systems solutions."

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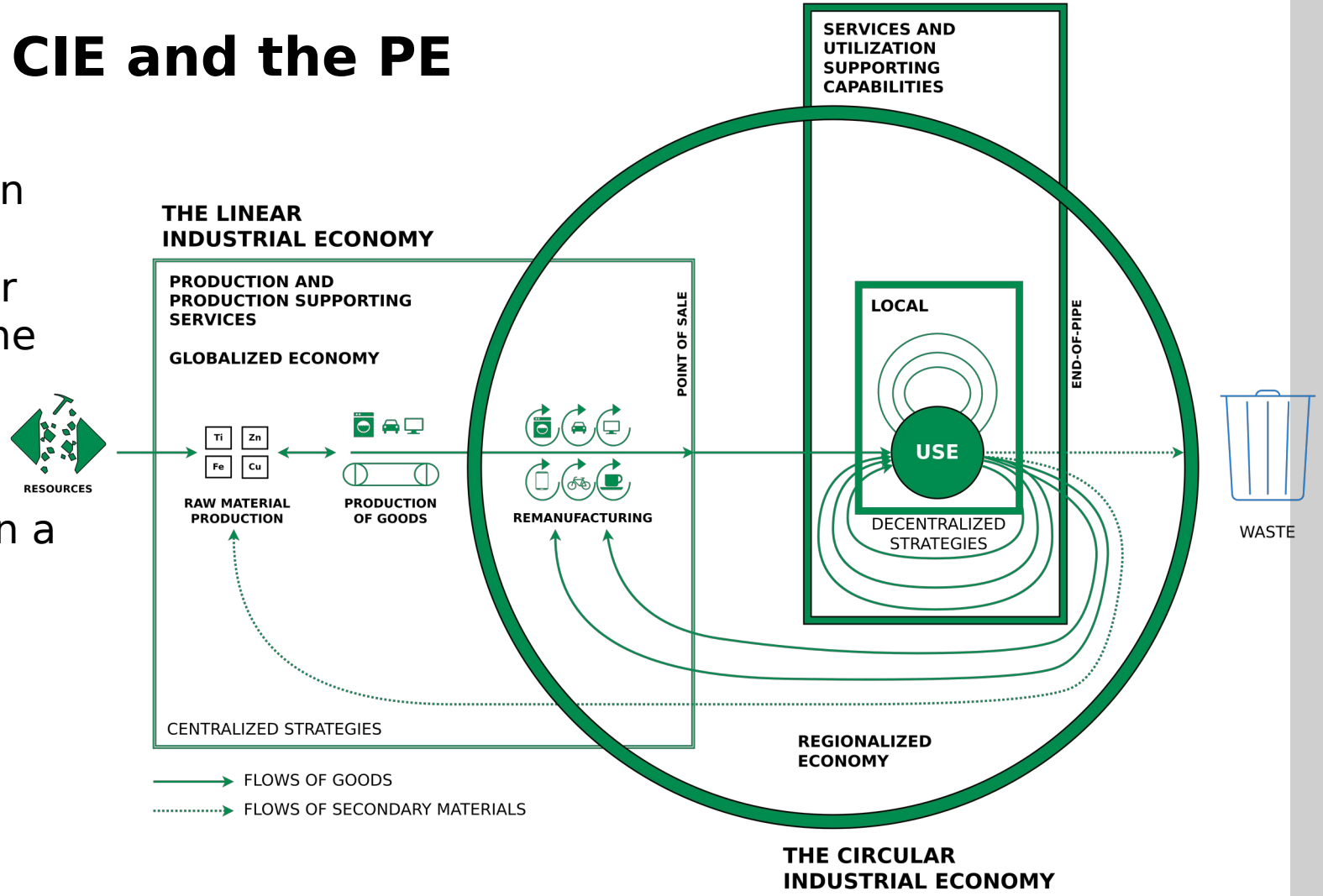
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  - "It maximises the profit potential by exploiting sufficiency, efficiency and systems solutions."
  - "In addition, by maintaining the ownership of objects and embodied resources, it creates long-term corporate and national resource security at low cost."

Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Situating the LIE, the CIE and the PE

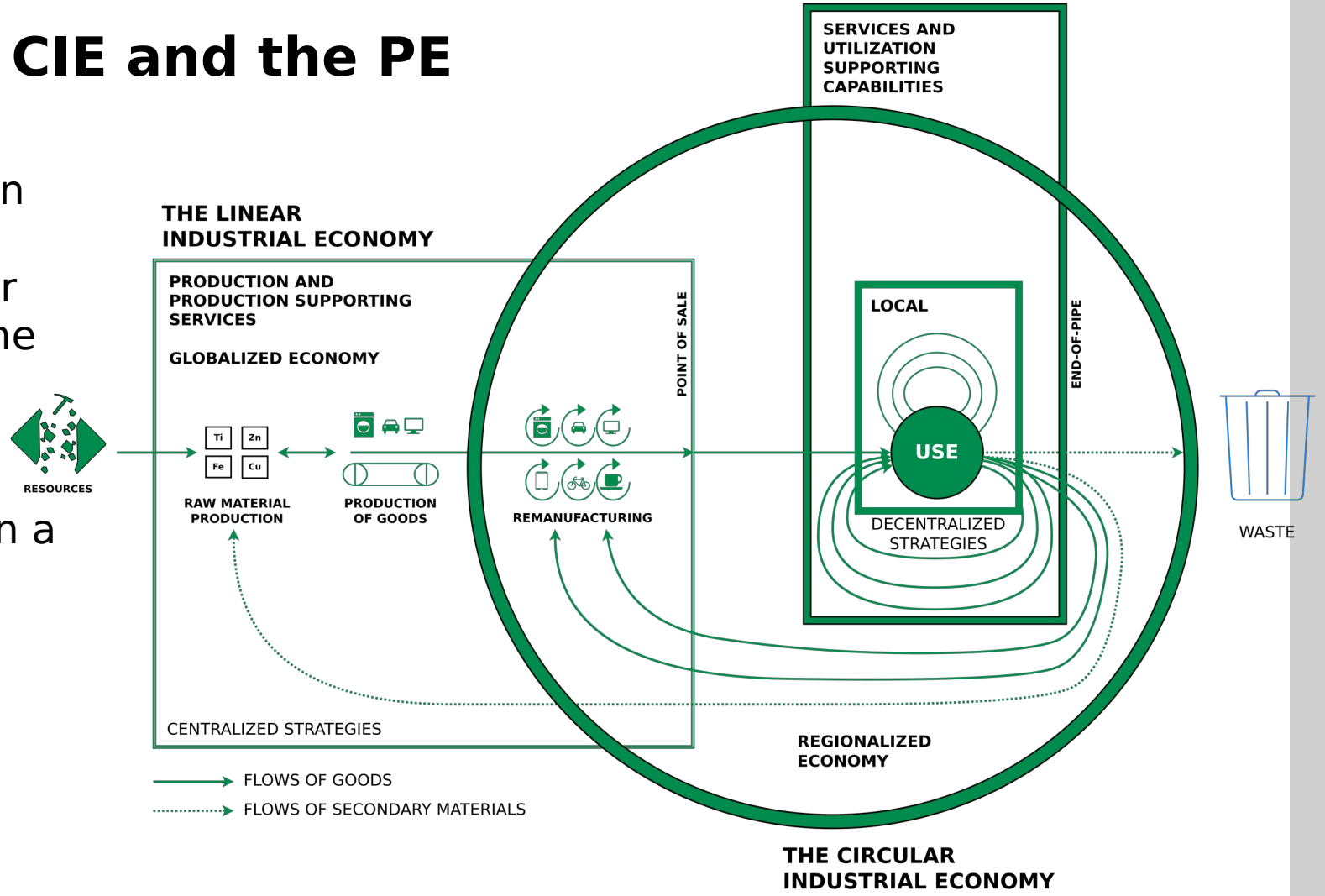
- **Circle:** Managing the utilisation or use phase of stocks of manufactured objects and their components, by maintaining the value and quality of infrastructure, buildings, investment goods, equipment and durable consumer goods in a local or regional economy



Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Situating the LIE, the CIE and the PE

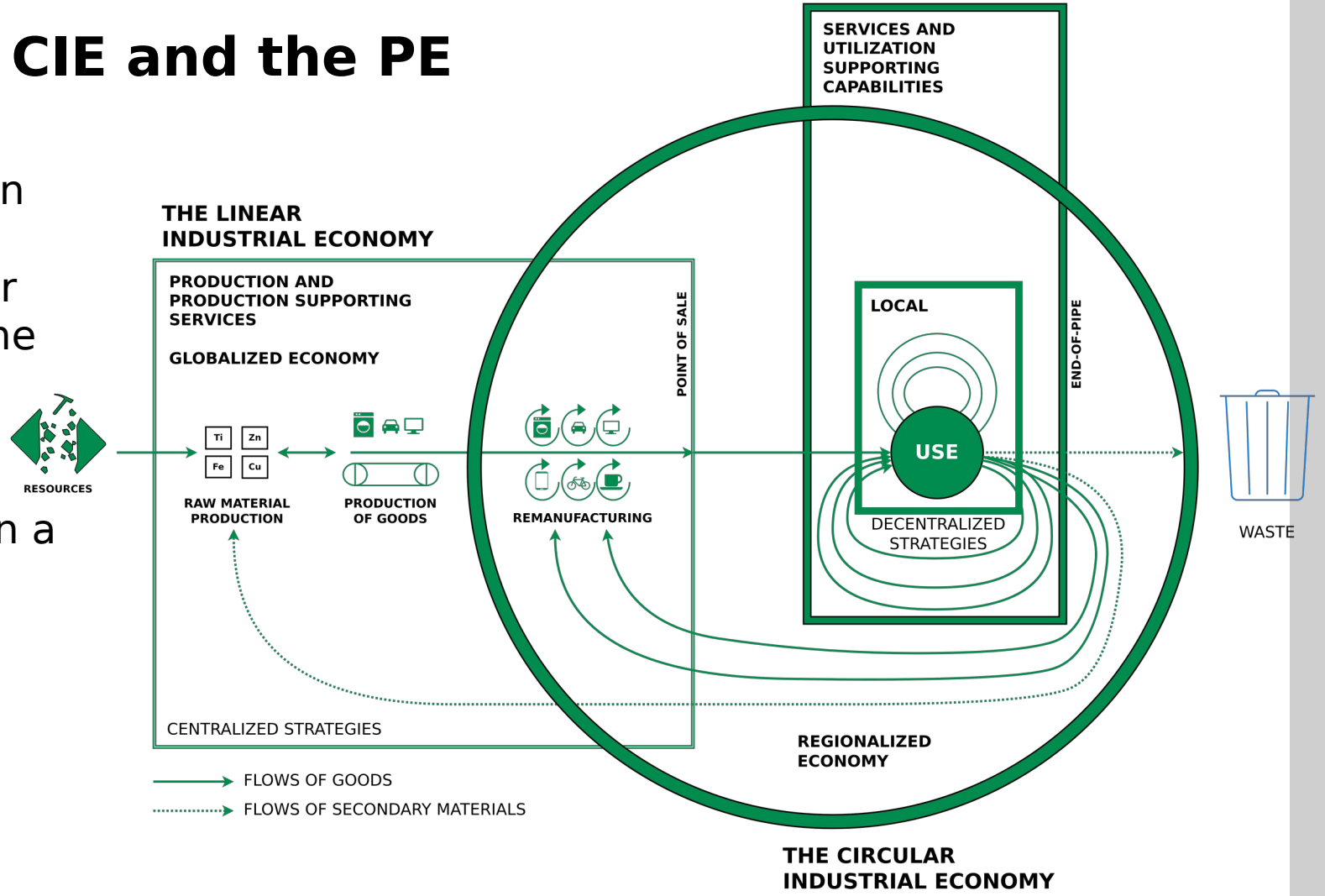
- **Circle:** Managing the utilisation or use phase of stocks of manufactured objects and their components, by maintaining the value and quality of infrastructure, buildings, investment goods, equipment and durable consumer goods in a local or regional economy
- **Small square:** Local use-focused PE



Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

Situating the LIE, the CIE and the PE

- **Circle:** Managing the utilisation or use phase of stocks of manufactured objects and their components, by maintaining the value and quality of infrastructure, buildings, investment goods, equipment and durable consumer goods in a local or regional economy
- **Small square:** Local use-focused PE
- **Big square:** Flows of used materials returning to the raw material producer to recover molecules and atoms in a globalised economy



Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

CONCLUSION

Conclusion

- Linear Economy (LE) → “Take – make – dispose”
- Circular Economy (CE) → Maintain value and utility as high as possible for as long as possible
- Performance Economy → Sell results instead of objects

Additional Resources

- Walter R. Stahel (2019) – The Circular Economy: A User's Guide
- Website of the Ellen MacArthur Foundation – Link

Questions?